[1994ApJ...424..507B]

THE ASTROPHYSICAL JOURNAL, 424:507–509, 1994 March 20
© 1994. The American Astronomical Society. All rights reserved. Printed in U.S.A.

LABORATORY MEASUREMENTS OF ROTATIONAL TRANSITIONS OF LITHIUM HYDRIDE IN THE FAR-INFRARED

M. BELLINI, P. DE NATALE, AND M. INGUSCIO
European Laboratory for Non-Linear Spectroscopy (LENS) and Department of Physics of the University, Largo E. Fermi 2, I-50125 Firenze, Italy
e-mail: ir@ifilens.bitnet

AND

E. FINK, D. GALLI, AND F. PALLA
Astrofisico di Arcetri, Largo E. Fermi, I-50125 Firenze, Italy
e-mail: fink@arcetri.astro.it; galli@arcetri.astro.it;
palla@arcetri.astro.it
Received 1993 August 5; accepted 1993 September 28

ABSTRACT

We report laboratory frequency measurements of 14 pure rotational transitions of ^{7}LiH and ^{6}LiH in the fundamental and in the first two excited vibrational states. Data have been taken with a tunable far-infrared spectrometer in the 400–3000 GHz frequency range. We also discuss the astrophysical and cosmological relevance of LiH.

Subject headings: infrared: ISM: lines and bands — line: identification — molecular data

1. INTRODUCTION

Lithium hydride, the simplest neutral heteronuclear diatomic molecule, has attracted for many years the interest of theoretical and experimental spectroscopists (e.g., Vidal & Stwalley 1982; Plummer, Herbst, & De Lucia 1984). The small number of electrons in the molecule allows precise ab initio calculations of its structure and makes LiH an ideal workbench for tests of theoretical models, as in the case of the Born-Oppenheimer approximation, which is no longer valid for such a light molecule. In addition, lithium hydride is potentially interesting for the astrophysical implications (Kirby & Dalgarno 1978). In particular, a search for roto-vibrational lines emitted at high redshifts has been reported recently by de Bernardis et al. (1993).

We are currently investigating systematically in the laboratory the pure rotational spectrum of LiH and its isotopic substitutes, relying on accuracies better than 100 kHz for most of the measured line frequencies. Values of the collisional parameters have been presented in De Natale et al. (1993), while an extensive fit of the molecular constants of all the stable isotopic substitutes will be presented elsewhere (Bellini et al. 1993). In the present paper we report the laboratory measured frequencies of 14 purely rotational transitions of ^{7}LiH and ^{6}LiH in the fundamental and in the first two excited vibrational states, taken with a tunable far-infrared (FIR) spectrometer in the 400–3000 GHz frequency range. Present values are about two orders of magnitude more accurate than the frequencies measured by Maki, Olson, & Thompson (1990). Accuracies comparable to the present ones have been achieved with microwave techniques by Plummer et al. (1984), limited only to the lowest rotational line ($J = 0 \rightarrow 1$) of $v = 0$ and 1.

2. EXPERIMENTAL RESULTS

Data were taken with the tunable far-infrared spectrometer (TuFIR) at LENS, in Firenze, Italy. The spectrometer generates coherent FIR radiation in the 300–5500 GHz spectral range by nonlinear mixing of radiation from two frequency-stabilized ($\nu/\Delta\nu \sim 10^9$) CO_2 lasers and a microwave synthesizer (HP 8341B) onto a metal-insulator-metal (MIM) diode. FIR radiation, emitted from the diode, is collimated by an off-axis parabolic mirror and after passing through the absorption cell is phase-sensitively detected onto a Si-composite bolometer operated at 1.4 K. The uncertainty in the FIR radiation frequency is determined by the combined uncertainty in the frequency lock of the lasers and is estimated to be around 35 KHz. A more detailed description of the apparatus can be found elsewhere (e.g. Zink et al. 1990).

The gaseous LiH sample was produced in a heat pipe cell consisting of a 120 cm long stainless steel tube with 25 mm inside diameter. The central 90 cm of the tube were heated to about 500–600 C by means of a tightly wound heating wire. The end sections of the pipe were cooled with circulating water and fitted with two high-density polyethylene windows, transparent to FIR radiation. Both LiH powder and metallic Li in an atmosphere of H_2 were used in the cell without significant variations in the production rate of the gaseous sample. ^{6}Li was present in terrestrial isotopic abundance (^{6}Li/^{7}Li $\simeq$ 0.08) both in the powder and in the metallic wire. Pressures in the cell were kept low enough, 0.5–1.0 torr, to limit possible pressure shifts and, mainly, collisional broadening of the lines, while still having good signals.

A total of 14 pure rotational transition lines were recorded for ^{7}LiH and ^{6}LiH at several values of the vibrational quantum number ($v = 0$, 1, 2 for ^{7}Li, and $v = 0$, 1 for ^{6}Li). Good signal-to-noise (S/N) ratios ($\sim$20) were generally obtained for ^{7}LiH in the vibrational ground state, while poorer S/N values resulted for ^{6}LiH and for recordings relative to excited vibrational levels. To partly compensate for the lower thermal population of excited levels, higher temperatures were usually adopted in such recordings. About five scans were averaged for each line and spectra were fitted to a Voigt profile by means of a computer program kindly provided by K. Chance of the Harvard-Smithsonian Center for Astrophysics. An example of a typical spectrum with the computed fit is shown in Figure 1 for the $J = 4 \rightarrow 5$, $v = 0$ transition of ^{7}LiH. The measured frequencies are listed in Tables 1 and 2, together with their 1 σ uncertainty, for ^{7}LiH and ^{6}LiH, respectively.

Although S/N ratios were generally more than acceptable

© American Astronomical Society • Provided by the NASA Astrophysics Data System

[1994ApJ...424..507B]

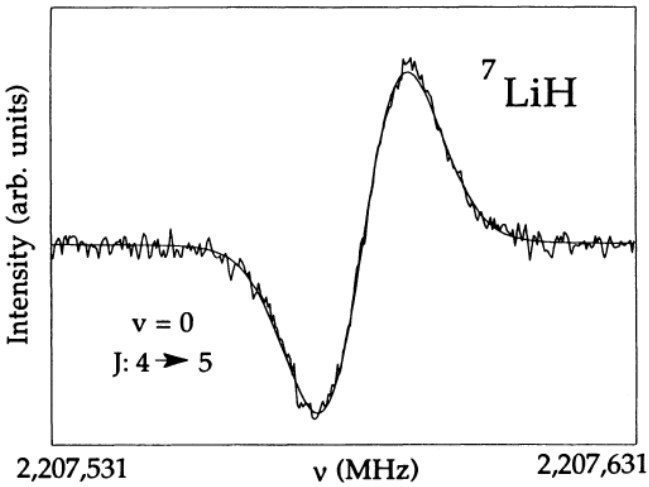


FIG. 1.—Experimental recording of the $J = 4 \rightarrow 5$ transition of ^{7}LiH in the fundamental vibrational mode. The fit is also shown.

for many of the lines under study, in some cases there were problems of repeatability for frequencies in subsequent scans. Such problems were probably due to fluctuations in pressure and temperature (giving rise to fluctuations in LiH density) which slightly changed shape and intensity of the absorbing line during any single scan. The resulting deformations of the lines can easily account for the observed spread in central frequencies. The values reported in Tables 1 and 2 are weighted averages of several scans performed on each line. For some lines, we had recordings at different pressures, and the adopted frequency was the zero pressure extrapolation based on a linear fit.

TABLE 1

MEASURED ROTATIONAL FREQUENCIES OF ^{7}LiH

v	J'–J''	ν (MHz)[a]
0.....	0–1	443,952.93 (12)
0.....	2–3	1,329,415.425 (50)
0.....	3–4	1,769,711.222 (84)
0.....	4–5	2,207,584.583 (55)
0.....	6–7	3,073,719.689 (46)
1.....	4–5	2,144,133.66 (14)
1.....	5–6	2,566,428.32 (30)
2.....	2–3	1,253,771.89 (18)
2.....	5–6	2,491,759.62 (14)

[a] Numbers in parentheses are 1 σ uncertainties and refer to the last digits.

TABLE 2

MEASURED ROTATIONAL FREQUENCIES OF ^{6}LiH

v	J'–J''	ν (MHz)[a]
0.....	3–4	1,806,284.25 (13)
0.....	4–5	2,253,109.84 (15)
0.....	5–6	2,696,797.93 (12)
1.....	2–3	1,317,579.78 (30)
1.....	5–6	2,618,408.03 (15)

[a] Numbers in parentheses are 1 σ uncertainties and refer to the last digits.

3. DISCUSSION

From the astrophysical viewpoint, the interest for lithium hydride is considerable. Its detection in the interstellar medium would provide an independent estimate of the lithium abundance and put valuable constraints on its formation and destruction mechanisms in stars and on primordial nucleosynthesis models. To date, however, LiH has not been observed in space. Kirby & Dalgarno (1978) have estimated its abundance in diffuse clouds exposed to the interstellar radiation field and concluded that the destruction of LiH is quite rapid in the unshielded field. The main production mechanisms are via radiative association of Li^+ with H_2 molecules, followed by dissociative recombination, or by neutralization of Li^+ on dust grains. In both cases, these channels are quite slow, and the predicted column densities, $\sim 10^6$ cm^{-2}, too small to allow detection through visual absorption. Bochkarev & Khersonskii (1985) have suggested that in regions of higher gas density, such as the dense cores observed in NH_3 or CS (e.g., Benson & Myers 1989), the interstellar radiation field is effectively attenuated, and conditions are more favorable for an enhancement of the expected abundance. The cool, neutral winds from pre–main-sequence T Tauri stars appear also as suitable places for the production of LiH molecules at significant rates. However, the full chemical network in both dense cores and stellar winds requires the knowledge of the rates of important reactions, such as $LiH + H \rightarrow Li + H_2$, which are not known at present. It would be important to reconsider the problem of LiH abundance, particularly since most of the measured rotational transitions listed in Table 1 are accessible to the long wavelength spectrometer (LWS, 45–200 mm) onboard *ISO* satellite.

In a completely different context, the production of LiH has been considered during the postrecombination epoch in the early universe (Dubrovich 1977; Lepp & Shull 1984; Puy et al. 1993). The presence of this trace molecule in the primordial gas is of importance as a cooling agent, to maintain the temperature low enough to favor the formation of the first structures. In the dust-free primordial gas the formation of LiH is thought to occur via radiative association,

$$\mathrm{Li} + \mathrm{H} \rightarrow \mathrm{LiH} + h\nu\,, \qquad (1)$$

with an estimated rate coefficient of about 10^{-17} cm^3 s^{-1}. Destruction processes are dominated by the cosmic background radiation, which limits the formation epoch at redshifts $z_{\mathrm{form}} \ll z_{\mathrm{rec}} \simeq 1000$. Depending on the cosmological model, the predicted abundances of LiH are in the range 10^{-13}–10^{-11} and $z_{\mathrm{form}} \simeq 200$ (Lepp & Shull 1984). More recently, Dalgarno & Lepp (1987) have argued that the formation of the molecular ion LiH^+, rather than the neutral hydride, should be favored in the primordial chemistry. Since lithium remains ionized to lower radiation temperature than hydrogen, the reaction

$$\mathrm{Li}^+ + \mathrm{H} \rightarrow \mathrm{LiH} + h\nu\,, \qquad (2)$$

should take place instead of reaction (1). However, two destruction mechanisms, photodestruction by the cosmic background radiation and dissociative recombination, may severely limit its importance and keep the abundance of LiH^+ well below that of LiH. Unfortunately, photodissociation cross sections for LiH^+ have not been calculated. As for the other process, $LiH^+ + e \rightarrow Li + H$, the potential energy curves are suitably located for the reaction to occur (Stwalley & Zemke

© American Astronomical Society • Provided by the NASA Astrophysics Data System

[1994ApJ...424..507B]

1993). Lacking a detailed calculation of the dissociative recombination cross sections, we have estimated from preliminary model calculations (Fink, Galli, & Palla 1993) that even with a reaction rate of $k = 10^{-11}\ T^{-0.5}\ \mathrm{cm}^{-3}\ \mathrm{s}^{-1}$, quite small compared to typical rates (e.g., Mitchell 1990), the predicted abundance of LiH^+ through reaction (2) will be lower than that of LiH. For $k = 10^{-8}\ T^{-0.5}\ \mathrm{cm}^3\ \mathrm{s}^{-1}$, we find a ratio $LiH^+/LiH \simeq 10^{-3.5}$ at $z = 180$. Thus, we conclude that it is very likely that the primordial chemical network did in fact yield a trace abundance of neutral lithium hydride. If $z_{\mathrm{form}} = 180$, as in Lepp & Shull (1984), then the redshifted roto-vibrational lines would fall in the atmospheric window around 1.5 mm.

This conclusion is important in the light of the recent search at these wavelengths for LiH lines at high redshifts by de Bernardis et al. (1993). Specifically, these authors report a 95% upper limit of $\sim$20 mK for the intensity of three lines of the first two roto-vibrational branches, showing that sensitive studies of primordial molecules can be carried out with current instrumentation. Obviously, more observations are needed to confirm these initial results.

It is important to note that, given the excitation conditions expected at these redshifts, lines from pure rotation transitions in the fundamental vibrational level are stronger than the roto-vibrational transitions considered by de Bernardis et al. (1993). The prospects of observing these lines redshifted at centimeter wavelengths should be carefully considered. Finally, a correct identification of weak features in noisy spectra requires the proper knowledge of the redshift of molecule formation. Recent analyses have shown that the appropriate z_{form} is between 350 and 400, rather than 180 (Puy et al. 1993; Fink et al. 1993). In this case, redshifted roto-vibrational lines of LiH might be detectable also in the atmospheric window around 3 mm.

It is a pleasure to acknowledge useful discussions with Prof. R. Moccia and Prof. A. E. Glassgold.

REFERENCES

Bellini, M., Brown, J. M., De Natale, P., Varberg, T., & Inguscio, M. 1993, in Proc. 13th Colloq. on High-Resolution Molecular Spectroscopy, ed. E. Canè, F. Tullini, & E. Venuti (Bologna: CUSL), in press
Benson, P. J., & Myers, P. C. 1989, ApJS, 79, 89
Bochkarev, N. G., & Khersonskii, V. K. 1985, Bull. Special Astrophys. Obs. North Caucasus, 18, 84
Dalgarno, A., & Lepp, S. 1987, in Astrochemistry, IAU Symp. 120, ed. M. S. Vardya & S. P. Tarafdar (Dordrecht: Reidel), 109
de Bernardis, P., et al. 1993, A&A, 269, 1
De Natale, P., Bellini, M., Inguscio, M., Buffa, G., & Tarrini, O. 1993, J. Molec. Spectrosc., in press
Dubrovich, V. K. 1977, Soviet Astron. Lett., 3, 128
Fink, E., Galli, D., & Palla, F. 1993, in preparation
Kirby, K., & Dalgarno, A. 1978, ApJ, 224, 444
Lepp, S., & Shull, J. M. 1984, ApJ, 280, 465
Maki, A. G., Olson, W. B., & Thompson, G. 1990, J. Molec. Spectrosc., 144, 257
Mitchell, J. B. A. 1990, Phys. Rep., 186, 215
Plummer, G. M., Herbst, E., & De Lucia, F. C. 1984, J. Chem. Phys., 81, 4893
Puy, D., Alecian, G., Le Bourlot, J., Lorat, J., & Pineau des Forts, G. 1993, A&A, 267, 337
Stwalley, W. C., & Zemke, W. T. 1993, J. Phys. Chem. Ref. Data, 22, 87
Vidal, C. R., & Stwalley, W. C. 1982, J. Chem. Phys., 77, 883
Zink, L. R., De Natale, P., Pavone, F. S., Prevedelli, M., Evenson, K. M., & Inguscio, M. 1990, J. Molec. Spectrosc., 143, 304

© American Astronomical Society • Provided by the NASA Astrophysics Data System